1. **Define cloud computing according to the NIST definition. Mention two advantages of cloud computing compared to traditional computing.**

According to the **National Institute of Standards and Technology (NIST)**, **Cloud Computing** is defined as:
"A model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction."

Two advantages of cloud computing compared to traditional computing:

- **Scalability and Flexibility:**
  Cloud resources can be easily scaled up or down based on user demand, unlike traditional systems that require manual hardware upgrades.
- **Cost Efficiency:**
  Cloud computing follows a *pay-as-you-go* model, reducing the need for large upfront investments in hardware and maintenance costs associated with traditional infrastructure.

2. **State two key features of Google App Engine.**

Two key features of **Google App Engine** are:

- **Automatic Scaling:**
  Google App Engine automatically scales applications up or down based on the amount of traffic or requests, ensuring optimal performance without manual intervention.
- **Managed Infrastructure:**
  It provides a fully managed environment where Google handles server management, load balancing, patching, and monitoring, allowing developers to focus only on writing and deploying code.

3. **What is the Open Virtualization Format (OVF)?**

The **Open Virtualization Format (OVF)** is an **open standard** for packaging and distributing **virtual appliances or virtual machines**. It defines a portable and efficient method to describe the contents of a virtual machine, including its configuration, disk images, and metadata, in a platform-independent manner.

4. **Explain how cloud systems can be integrated with wireless sensor networks and list two applications.**

Integration of Cloud Systems with Wireless Sensor Networks (WSNs):
Cloud systems can be integrated with Wireless Sensor Networks by connecting sensor nodes (which collect data such as temperature, humidity, or motion) to the cloud through gateways. The gateway aggregates sensor data and transmits it to the cloud over the Internet, where it is stored, processed, and analyzed using cloud-based

applications and services. This integration enables real-time data monitoring, large-scale analytics, and remote accessibility.

Two applications of Cloud–WSN integration:

- **Smart Agriculture:**
  Sensors monitor soil moisture, temperature, and crop health, and the data is analyzed in the cloud to optimize irrigation and farming practices.
- **Smart Healthcare:**
  Wearable or environmental sensors collect patient health data, which is transmitted to the cloud for continuous monitoring and medical analysis by healthcare professionals.

5. Amazon EC2, Amazon S3, and Amazon SimpleDB and describe practical use-cases.

- **Amazon EC2 (Elastic Compute Cloud):**

Amazon EC2 provides scalable virtual servers (instances) in the cloud, allowing users to run applications on demand without investing in physical hardware.

Practical Use-Case:

Hosting web applications or running high-performance computing tasks such as data analysis, simulations, or AI model training.

- **Amazon S3 (Simple Storage Service):**

Amazon S3 offers secure, scalable, and durable object storage for any type of data, such as files, images, and backups.

Practical Use-Case:

Storing and retrieving large volumes of data like website images, backup files, or multimedia content for streaming platforms.

- **Amazon SimpleDB:**

Amazon SimpleDB is a NoSQL database service that provides simple and flexible data storage and querying capabilities for smaller, non-relational datasets.

Practical Use-Case:

Managing lightweight structured data, such as user profiles, product catalogs, or configuration settings for web applications.